

LENS: A Linked Analytics Workflow for Evaluating Course Redesign Using Fixed-Effects Models and Structural Equation Modeling

Abstract—Online course redesign is increasingly used to improve student success in online and hybrid learning, but evaluation remains difficult because course design records, implementation data, student survey responses, and academic outcomes are often stored separately. This study introduces LENS, a linked analytics workflow for evaluating course redesign using fixed-effects models and structural equation modeling. We analyze a de-identified institutional dataset collected at a public university from early 2024 through the first half of 2026, containing 1,568 student-course observations across 19 courses, 67 course-term cells, and 268 section-term rows. The workflow links QM-style course review data summarized through eight Legon-aligned redesign clusters, student perceptions from the 30-item Online Learning Self-Efficacy (OLSE) survey, and final numeric grades. LENS combines course and term fixed-effects grade models, OLSE reliability analysis and ordinal confirmatory factor analysis, and SEM-based mechanism-oriented analysis of redesign–self-efficacy–grade pathways. Results show that post-redesign offerings were associated with a 0.380-point increase in final numeric grade, with stable estimates across standardized, instructor-adjusted, course-term aggregate, CR2/Satterthwaite small-cluster, and leave-one-course-out specifications. OLSE showed high overall reliability ($\alpha = 0.932$), and ordinal CFA supported a coherent higher-order OLSE construct, with five-domain and single-factor models serving as supporting checks, and SEM sensitivity analyses indicated positive indirect redesign–OLSE–grade associations across latent, manifest, and redesign-intensity specifications. These findings suggest that LENS can support privacy-aware, mechanism-oriented course redesign evaluation by linking design changes, student self-efficacy, and academic outcomes in a reproducible institutional analytics workflow.

Index Terms—course redesign, learning analytics, linked data, fixed-effects models, structural equation modeling, online learning self-efficacy

I. INTRODUCTION

Course redesign has become an important strategy for improving the quality, accessibility, and effectiveness of higher education. As institutions expand online and hybrid offerings, instructional design teams increasingly rely on structured course-quality frameworks, such as Quality Matters (QM), to improve course organization, learner support, assessment alignment, interaction, and technology use [7], [9], [12]. These efforts are motivated by practical goals: improving student outcomes, reducing course-level barriers, and supporting students in online learning environments [10], [11], [14].

In this study, Online Learning Self-Efficacy (OLSE) is treated as a theory-guided mechanism-oriented construct because self-efficacy is associated with motivation, persistence,

and performance [15], [16]. OLSE is measured through a student self-report survey that asks students how confident they are that they can perform specific online learning tasks [17]. Thus, OLSE captures students’ perceived online learning capabilities rather than direct behavioral activity or instructor judgments.

The central problem is that course redesign evaluation is frequently fragmented. Student grade outcomes, course-section records, survey responses, and redesign rubric data are often analyzed separately. As a result, institutions may report whether grades changed after redesign, but have less evidence about how redesign features relate to student self-efficacy and academic outcomes. This limits the practical value of analytics for instructional design teams because it does not clearly indicate which redesign dimensions may be connected to student confidence, engagement, or performance [3], [10].

Prior research has examined online course quality, self-efficacy, and student outcomes, but there remains a gap in reproducible data science workflows that integrate these elements into a single institutional analytics pipeline [1], [2], [5]. Existing studies often evaluate individual courses, rely on survey-only analyses, or examine student outcomes without linking them to structured redesign features [10], [11], [13]. Conversely, learning analytics dashboards may summarize performance patterns but often do not model latent constructs such as self-efficacy [3], [6].

This paper presents LENS, a Linked Analytics Workflow for Evaluating Course Redesign Using Fixed-Effects Models and Structural Equation Modeling. LENS provides a reproducible linked-data workflow for integrating course redesign records, course-term implementation data, student OLSE responses, and academic outcomes. The central argument is that course redesign evaluation should move beyond isolated pre/post grade comparisons by linking design changes to theory-guided student self-efficacy measures and grade outcomes within a single analytic pipeline. The workflow is intended to support mechanism-oriented institutional decision-making under realistic observational-data constraints. This argument leads to three research questions:

- **RQ1:** Are post-redesign course offerings associated with higher final numeric grades after accounting for course and term differences?
- **RQ2:** Does OLSE provide a reliable higher-order measurement structure suitable for mechanism-oriented modeling?

- **RQ3:** Are redesign exposure and redesign dimensions associated with final numeric grades through indirect OLSE-based associations?

These questions correspond to the three analytic layers of LENS: fixed-effects grade models for RQ1, OLSE reliability and confirmatory factor analysis for RQ2, and SEM-based mechanism-oriented modeling for RQ3. In summary, the paper contributes: (i) a linked institutional data architecture for course redesign evaluation; (ii) fixed-effects and robustness procedures for reporting adjusted grade associations; (iii) OLSE measurement checks using reliability analysis, five-domain CFA, higher-order CFA, and single-factor CFA; and (iv) SEM-based sensitivity analyses for examining redesign–OLSE–grade associations. Although this study is situated in online course redesign, LENS addresses a broader applied data-science problem: evaluating interventions when design records, implementation metadata, survey-based latent constructs, and administrative outcomes are stored in separate systems. The workflow contributes a reusable pattern for linking heterogeneous institutional data layers into interpretable, privacy-aware, mechanism-oriented evaluation outputs. The redesign variables are derived from QM-style course evaluation records, where item-level course review scores are summarized into eight Legon-aligned redesign clusters and then grouped into theory-guided redesign composites [7], [8]. We apply the workflow to a de-identified institutional dataset collected from real student-course observations at a public university from early 2024 through the first half of 2026. The dataset contains 1,568 student-course observations across 19 courses. Section II describes the linked data architecture, Section III presents the methodology, Section IV reports results, Section V discusses implications and limitations, and Section VI concludes.

II. LINKED DATA ARCHITECTURE

The LENS workflow uses a linked data architecture that connects course redesign information, course-term implementation records, student survey responses, and academic outcomes. This architecture is central because learning analytics depends on integrating data about learners, learning processes, learning environments, and institutional outcomes [1], [4], [5]. Without linking these sources, redesign evaluation is limited to either descriptive redesign summaries or outcome comparisons that do not account for implemented design features.

The architecture integrates three data layers. The student-course layer contains one row per student-course observation, including anonymized student identifiers, course identifiers, term and section identifiers, OLSE survey responses, and final numeric grades. The course-term layer describes each course offering using course, term, section, section-level instructor, and redesign-status identifiers. Course-redesign records are treated as course-level design artifacts associated with an instructor team during the redesign/evaluation period, while course-term and student-course records retain section-level instructor identifiers. The course-redesign layer contains course-level redesign information derived from QM-style faculty self-review and course evaluation records. These records include

TABLE I: Linked data layers used in the LENS workflow.

Data layer	Unit	Key fields	Role
Student-course	Student-course observation	Student ID, course, term, section, OLSE, grade	Grade and outcome OLSE indicators
Course-term	Course-section-term offering	Course, term, section, section-level instructor, redesign status	Timing and implementation
Course-redesign	Course	QM review scores, Legon clusters, composites	Design features

item-level standards with points possible, points awarded, result status, evidence, and suggestions for improvement. Item-level scores are aggregated into eight Legon-aligned redesign clusters: clarity of purpose, ease of use, course alignment, learner engagement, accessibility, knowledge acquisition, compliance, and learner support [7], [8].

The layers are joined using anonymized course, term, and section identifiers. The resulting analytic file contains 1,568 student-course observations across 19 courses, 67 course-term cells, and 268 section-term rows collected from early 2024 through the first half of 2026. The three linked backend datasets were treated as the final analytic inputs: a student-course dataset, a course-term implementation dataset, and a course-redesign dataset. The final analytic file contained complete grade, redesign-status, course, term, section, and OLSE item information for 1,568 student-course observations. Therefore, the analyses focus on the linked analytic sample rather than modeling survey nonresponse or raw administrative-record attrition. For RQ1, the linked file identifies pre- and post-redesign offerings within the same courses. For RQ2, it provides OLSE survey indicators for reliability and measurement analysis. For RQ3, it links redesign dimensions to OLSE and final grades for SEM-based mechanism-oriented analysis.

Redesign features were computed from course review records before being linked to student-course outcomes. Final numeric grades were not used to construct Legon cluster scores, redesign composites, OLSE measures, or redesign-intensity features. The linked analytic file was then used for modeling, with final numeric grade reserved as the outcome variable.

Because the dataset contains real student-course records and survey responses, all student, course, section, and instructor identifiers are de-identified before analysis. Learning analytics can raise privacy, transparency, and governance concerns, so the workflow is intended for course-level and program-level evaluation rather than individual-level prediction or automated decision-making [44]–[47]. Public reproducibility materials should include code, data dictionaries, masked or synthetic example data, and result-generation scripts rather than raw student-level records [42], [43].

Final numeric grade was treated as a continuous institutional outcome, and OLSE items were treated as ordered categorical indicators in the CFA models. All 30 OLSE items were observed for the analytic sample, so item-level OLSE missingness was 0% for every indicator.

OLSE survey responses were collected during the course offering before final numeric grades were finalized.

III. METHODOLOGY

Building on the linked data architecture described in Section II, the methodology is organized around the three research questions. The study uses a repeated cross-sectional course-term design in which student-course observations are linked to course-term implementation records and course-level redesign features. Because the same students are not followed longitudinally, the analysis estimates associations across course offerings rather than within-student changes over time [21], [22]. All analyses were conducted in R.

Fixed-effects models were estimated using linear regression specifications with course and term indicators, with additional small-cluster and leave-one-course-out checks for robustness. OLSE CFA and SEM models were estimated using lavaan; ordinal CFA models were estimated using WLSMV for ordered categorical indicators [30], [33]. Model fit was evaluated using CFI, TLI, RMSEA, and SRMR, and SEM results were interpreted as redesign-OLSE-grade associations.

A. Study Design

The repeated cross-sectional structure compares course offerings before and after redesign rather than following the same students over time. This design is appropriate for institutional course redesign evaluation because redesign is implemented at the course or course-section level, while grades and survey responses are observed at the student-course level. The unit of analysis for the primary analytic file is therefore the student-course observation.

This design supports three analytic tasks: outcome modeling through within-course pre/post comparisons, measurement modeling of OLSE survey responses, and mechanism analysis linking redesign exposure and redesign dimensions to grades through OLSE. The goal is not to build a predictive model for individual student risk classification, but to provide a reproducible workflow for course-level and program-level redesign evaluation [42], [43]. Accordingly, the empirical goal is stability and interpretability of linked analytics evidence rather than individual-level prediction or definitive causal identification.

B. Redesign Feature Engineering

Course redesign features were constructed from rubric-based pre-redesign and post-redesign review scores. The course review process produces item-level scores for QM-style standards, including points possible, points awarded, evidence, result status, and suggestions for improvement [7], [9]. These item-level review scores are aggregated into eight Legon-aligned redesign clusters that quantify the amount and type of course change [8].

The eight clusters are: Cluster 1, clarity of purpose; Cluster 2, ease of use; Cluster 3, course alignment; Cluster 4, learner engagement; Cluster 5, accessibility; Cluster 6, knowledge acquisition; Cluster 7, compliance; and Cluster 8, learner

TABLE II: Mapping from Legon’s clusters to redesign composites.

Composite	Legon’s clusters	Interpretation
Mastery redesign	1 clarity; 2 ease; 3 alignment; 6 knowledge	Clarity, alignment, objectives, feedback-oriented design
Social/vicarious redesign	4 engagement; 8 support	Interaction, communication, instructor presence, support
Structural/compliance redesign	5 accessibility; 7 compliance	Accessibility, policy clarity, navigation, organization

support. For each redesign cluster k , a normalized change score was computed as:

$$\Delta_k = \frac{Post_k}{Max_k} - \frac{Pre_k}{Max_k}, \quad (1)$$

where $Post_k$ is the post-redesign score, Pre_k is the pre-redesign score, and Max_k is the maximum possible score for that cluster. Positive values represent increases in cluster scores after redesign.

To reduce dimensionality and avoid overfitting from modeling all eight redesign clusters simultaneously, the cluster-level change scores were grouped into three theory-guided redesign composites. Table II summarizes the mapping from the eight Legon clusters to the composites used in the fixed-effects and SEM models [8], [10]. An overall redesign intensity score was also computed as the average normalized change across all eight clusters. Composite and intensity scores were used to reduce dimensionality in the grade and SEM models, while one-cluster-at-a-time specifications were retained as sensitivity checks to assess whether conclusions depended on the composite grouping.

C. Fixed-Effects Grade Model for RQ1

RQ1 asks whether post-redesign offerings are associated with higher final numeric grades after accounting for course and term differences. The primary model is:

$$Y_{ict} = \alpha_c + \gamma_t + \beta PostRedesign_{ct} + \varepsilon_{ict}, \quad (2)$$

where Y_{ict} is the final numeric grade for student i in course c during term t , α_c represents course fixed effects, γ_t represents term fixed effects, and $PostRedesign_{ct}$ indicates whether the offering occurred after redesign. Course fixed effects adjust for time-invariant differences across courses, while term fixed effects adjust for common temporal differences [21], [23]. Estimates are interpreted as adjusted associations because unobserved time-varying changes may coincide with redesign implementation [22], [24].

Additional specifications examined redesign intensity and redesign composites using interactions:

$$Y_{ict} = \alpha_c + \gamma_t + \beta_1 PostRedesign_{ct} + \beta_2 (PostRedesign_{ct} \times RedesignFeature_c) + \varepsilon_{ict}. \quad (3)$$

Course-level cluster-robust standard errors were used to account for within-course dependence among student-course observations. Because the number of courses was limited, inference was also evaluated using a CR2 small-sample cluster-robust correction with Satterthwaite degrees of freedom. In

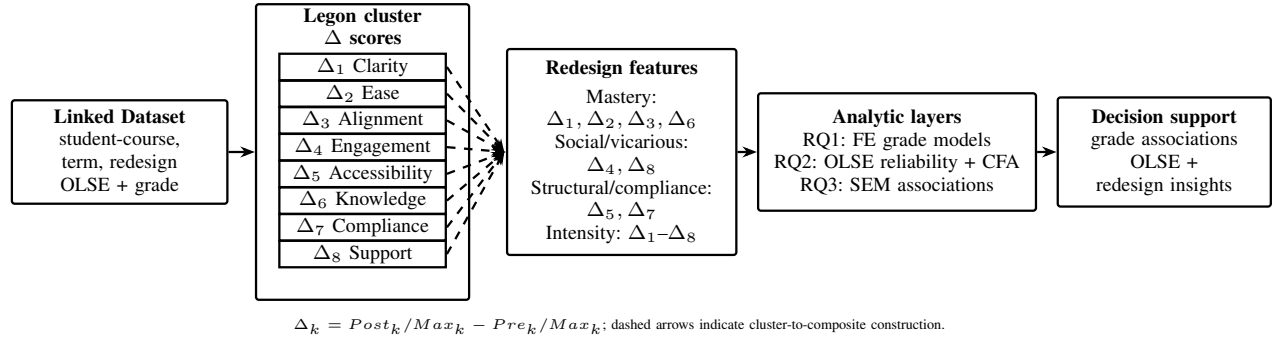


Fig. 1: Compact LENS workflow. Linked institutional records are transformed into redesign features, analyzed through FE, CFA, and SEM layers, and summarized as course-level decision-support outputs.

addition, we conducted a leave-one-course-out sensitivity analysis by re-estimating the primary fixed-effects model 19 times, each time excluding one course. These checks evaluate the stability of the adjusted grade association; causal interpretation is discussed in Section III-F.

D. OLSE Reliability and Confirmatory Factor Analysis for RQ2

RQ2 asks whether OLSE provides a reliable higher-order measurement structure suitable for use as a mechanism-oriented modelling. OLSE is a student self-report survey that captures students' perceived confidence in performing online learning tasks [17]. The instrument includes 30 ordinal survey items on a 1–5 Likert-type confidence scale, organized into five domains: online course completion, social interaction with classmates, use of course management system tools, instructor interaction, and academic interaction with classmates. Internal consistency was evaluated using Cronbach's alpha for each OLSE domain and for the overall 30-item scale [26]. A summary of the OLSE survey prompts is provided in Appendix A.

Ordinal CFA models were estimated to evaluate the latent structure of the survey. The five-domain correlated CFA was estimated as a measurement check reflecting the original OLSE domain organization. Because the domains were expected to reflect a broader online learning self-efficacy construct, a higher-order CFA was then estimated and used as the primary measurement basis for the mechanism model. As a supplemental robustness check, a single-factor CFA was also estimated to evaluate whether the measurement conclusion depended on the higher-order specification. Because OLSE items are ordinal Likert-type responses, WLSMV estimation was used for ordered categorical indicators [31]–[34]. Pre/post measurement comparability was evaluated using configural, metric, and threshold-plus-loading equality models [36]–[38]. For SEM, OLSE was represented using domain-level parcels corresponding to the first-order CFA domains [39].

E. SEM-Based Mechanism-Oriented Model for RQ3

RQ3 asks whether redesign exposure and redesign dimensions are associated with final numeric grades indirectly through OLSE. SEM was used as a mechanism-oriented

associational model linking observed redesign variables, latent OLSE, and final numeric grade [27], [28], [30]. The purpose was to evaluate whether the observed redesign–grade association was consistent with an OLSE-based mechanism, not to establish definitive causal mediation. Let M_{ict} denote latent OLSE, R_{ct} denote post-redesign status, and D_c denote a redesign composite:

$$M_{ict} = a_0 + a_1 R_{ct} + a_2 D_c + u_{ict}, \quad (4)$$

$$Y_{ict} = b_0 + b_1 M_{ict} + b_2 R_{ct} + b_3 D_c + e_{ict}. \quad (5)$$

The indirect association from redesign to grade through OLSE is computed as the product of the redesign-to-OLSE path and the OLSE-to-grade path [41]. Indirect estimates are interpreted as redesign–OLSE–grade associations; causal identification limits are discussed in Section III-F.

F. Causal Interpretation and Identification Limits

The LENS workflow is guided by a theory of change in which course redesign may influence OLSE and, through OLSE, academic performance. However, the empirical design does not identify causal effects or causal mediation effects. Redesign was not randomly assigned, the same students were not followed longitudinally across pre- and post-redesign offerings, and redesign exposure varied at the course or course-term level. Course and term fixed effects adjust for stable course differences and common temporal differences, but they do not remove possible time-varying confounding from instructor practice, grading policy, student composition, advising support, or institutional conditions [22], [24]. Similarly, SEM indirect associations are interpreted as redesign–OLSE–grade associations rather than causal mediation effects because stronger assumptions about assignment, temporal ordering, and unmeasured confounding would be required [40], [41]. Robustness checks assess stability of the associations, not causal identification.

IV. RESULTS

The results are presented in the same sequence as the research questions. Section IV-A describes the sample and treatment timing, Section IV-B addresses RQ1, Section IV-C addresses RQ2, and Section IV-D addresses RQ3.

TABLE III: Dataset summary and pre/post descriptive statistics.

Quantity	Value
Student-course observations	1,568
Courses	19
Course-term cells	67
Section-term rows	268
Pre-redesign observations	942
Post-redesign observations	626
Mean pre-redesign grade	2.456
Mean post-redesign grade	2.715
Mean pre-redesign OLSE	3.429
Mean post-redesign OLSE	3.597

TABLE IV: Fixed-effects grade model results.

Model	Est.	SE	<i>p</i>
Course + term FE	0.380	0.118	0.0048
Grade z-score outcome	0.478	0.149	0.0048
Course + term + section-instructor	0.387	0.119	0.0044
Course-term aggregate	0.380	0.123	0.0062

TABLE V: RQ1 Robustness Checks

Check	Main finding
CR2/Satterthwaite	$\beta = 0.380$, SE = 0.124, df = 15.44, $p = 0.0075$
Leave-one-course-out	All 19 estimates positive; β range = 0.337–0.438
Instructor-adjusted FE	$\beta = 0.387$, SE = 0.119, $p = 0.0044$
Course-term aggregate	$\beta = 0.380$, SE = 0.123, $p = 0.0062$

A. Descriptive Statistics and Treatment Timing

The linked analysis dataset contained 1,568 student-course observations across 19 courses, 67 course-term cells, and 268 section-term rows. Of these observations, 942 were from pre-redesign offerings and 626 were from post-redesign offerings. Mean final numeric grade increased from 2.456 to 2.715, a descriptive difference of 0.259 grade points. Mean OLSE increased from 3.429 to 3.597. All 30 OLSE items were complete for the analytic sample; item-level missingness was 0% for every OLSE indicator.

Treatment timing varied across terms, with early terms containing only pre-redesign observations, the final term containing only post-redesign observations, and middle terms containing both. These descriptive patterns are not used as causal evidence; they motivate the fixed-effects layer because simple pre/post comparisons would mix redesign timing with differences across courses and terms [21], [22].

B. RQ1: Post-Redesign Grade Associations

In the primary course and term fixed-effects model, post-redesign offerings were associated with a 0.380-point increase in final numeric grade (SE = 0.118, $p = 0.0048$). The corresponding standardized model estimated an association of 0.478 standard deviations. Estimates remained stable after adding instructor indicators and when aggregated to the course-term level.

Table V summarizes the main RQ1 robustness checks, including small-cluster inference and leave-one-course-out re-estimation.

Additional specifications showed positive interactions for mastery redesign (estimate = 0.138, $p = 0.0258$), so-

TABLE VI: OLSE reliability estimates.

Scale	Items	α
OLSE Domain Q1	8	0.785
OLSE Domain Q2	5	0.706
OLSE Domain Q3	6	0.738
OLSE Domain Q4	5	0.695
OLSE Domain Q5	6	0.743
Overall OLSE scale	30	0.932

cial/vicarious redesign (estimate = 0.180, $p = 0.0086$), and overall redesign intensity (estimate = 0.128, $p = 0.0145$); the structural/compliance interaction was positive and near the conventional significance threshold (estimate = 0.123, $p = 0.0543$). Overall, the results provide consistent evidence for RQ1 across outcome scales, aggregation levels, and robustness specifications.

As shown in Table V, the primary estimate remained stable under CR2/Satterthwaite small-cluster inference and was positive in all leave-one-course-out specifications. These checks suggest that the adjusted grade association was not driven by a single course or by reliance on conventional clustered standard errors.

C. RQ2: OLSE Measurement Structure

OLSE was measured as a 30-item student self-report survey on a 1–5 confidence scale [17]. Domain-level reliability ranged from borderline acceptable to acceptable, and the overall OLSE scale showed high internal consistency.

We first estimated a five-domain correlated CFA to verify that the OLSE items followed the expected domain organization [29]. The five-domain CFA showed excellent approximate fit, $\chi^2(395) = 442.746$, $p = .049$, CFI = 0.998, TLI = 0.998, RMSEA = 0.009, and SRMR = 0.019. Because the five domains were highly correlated, we then estimated a higher-order CFA representing OLSE as a general online learning self-efficacy construct. The higher-order CFA showed similar excellent approximate fit, $\chi^2(400) = 445.796$, $p = .057$, CFI = 0.998, TLI = 0.998, RMSEA = 0.009, and SRMR = 0.019. The higher-order model was retained as the primary measurement model because it preserves the theoretically defined OLSE domains while representing their shared general factor. Because OLSE items are ordinal Likert-type indicators, the CFA models used WLSMV estimation for ordered categorical data [31], [33]. Because fit indices can be sensitive to model structure and sample conditions, these results are interpreted as evidence that OLSE functions coherently as a general self-efficacy construct in this analytic sample, rather than as definitive evidence that all five first-order domains are empirically distinct [28], [35]. Standardized item loadings ranged from 0.578 to 0.654; higher-order loadings ranged from 0.985 to 0.998; and SEM parcel loadings ranged from 0.829 to 0.883. Pre/post comparability models were successfully estimated, with CFI = 1.000 and RMSEA = 0.000 across configural, metric, and threshold-plus-loading equality models [36]–[38]. Additional response-distribution diagnostics showed that all five ordered response categories were observed for every OLSE item in both pre- and post-redesign groups. Responses

TABLE VII: Primary SEM structural paths.

Path	Est.	SE	p
OLSE $\leftarrow$ post-redesign	0.421	0.080	< .001
OLSE $\leftarrow$ post $\times$ mastery	0.228	0.081	0.005
OLSE $\leftarrow$ post $\times$ social/vicarious	0.445	0.055	< .001
OLSE $\leftarrow$ post $\times$ structural/compliance	0.323	0.070	< .001
Grade $\leftarrow$ OLSE	0.373	0.032	< .001
Grade $\leftarrow$ post-redesign	0.197	0.049	< .001

TABLE VIII: Indirect associations through OLSE.

Indirect association	Est.	SE	p
Post $\rightarrow$ OLSE $\rightarrow$ grade	0.157	0.034	< .001
Mastery $\rightarrow$ OLSE $\rightarrow$ grade	0.085	0.034	0.012
Social/vicarious $\rightarrow$ OLSE $\rightarrow$ grade	0.166	0.022	< .001
Structural/compliance $\rightarrow$ OLSE $\rightarrow$ grade	0.120	0.028	< .001

were concentrated in the middle-to-high confidence categories, with a modest post-redesign shift toward higher confidence responses. This pattern supports the use of ordinal CFA while cautioning against overinterpretation of sparse low-confidence categories. As a supplemental robustness check, the single-factor CFA also showed strong fit, as summarized in Appendix Table XI. This supports representing OLSE as a coherent general self-efficacy construct while cautioning against overinterpreting domain-specific distinctions. Taken together, the reliability, CFA, response-distribution, and robustness results support the use of OLSE as a higher-order OLSE construct for RQ3, while cautioning against strong domain-specific interpretations.

D. RQ3: Redesign–OLSE–Grade Pathways

The primary SEM used five OLSE domain parcels as indicators of latent OLSE and showed strong fit: $\chi^2(25) = 30.081$, $p = .221$, CFI = 0.999, TLI = 0.999, RMSEA = 0.011, and SRMR = 0.009 [27], [28], [35]. Tables VII and VIII report structural paths and indirect associations.

Post-redesign status and all three redesign composites were positively associated with OLSE. OLSE was positively associated with final numeric grade (estimate = 0.373), and post-redesign status retained a positive direct association with grade (estimate = 0.197). The largest indirect estimate was observed for the social/vicarious pathway, suggesting that interaction, communication, instructor presence, and learner support may be especially relevant to the OLSE pathway [18], [20]. Overall, the SEM results are consistent with the proposed OLSE mechanism.

Sensitivity analyses showed that the mechanism-oriented pattern was stable across alternative SEM specifications. A manifest OLSE path model produced similar indirect associations for post-redesign status, mastery redesign, social/vicarious redesign, and structural/compliance redesign. An overall redesign-intensity SEM also showed positive indirect associations through OLSE for both post-redesign exposure and redesign intensity. These results suggest that the OLSE pathway interpretation was not dependent on a single latent-variable specification.

V. DISCUSSION

A. Overall Interpretation of Findings

The findings align with the three research questions that organize LENS. For RQ1, post-redesign offerings were associated with higher final numeric grades after accounting for course and term differences, and this association was stable across small-cluster and leave-one-course-out checks. For RQ2, the reliability, CFA, response-distribution, single-factor robustness, parcel-loading, and pre/post comparability results suggest that OLSE can be represented as a coherent higher-order construct suitable for mechanism-oriented analysis in this dataset. For RQ3, the SEM results indicate that redesign exposure and redesign dimensions were positively associated with grades through OLSE-consistent indirect associations.

The main contribution of LENS is that it moves course redesign evaluation beyond a single pre/post comparison. A grade-only analysis can indicate whether outcomes changed, but it cannot explain which design dimensions may be connected to student learning experiences. This aligns with prior arguments that learning analytics should support interpretation of learning processes rather than only reporting performance patterns [1], [3]. The robustness checks strengthen the stability of the workflow outputs, while the observational design motivates interpreting LENS as decision support.

B. Interpreting the Mechanism-Oriented Workflow

For RQ1, the stability of the fixed-effects estimates across raw-grade, standardized-grade, instructor-adjusted, course-term aggregate, CR2/Satterthwaite, and leave-one-course-out specifications suggests that the grade association was not dependent on a single outcome scale, inferential method, or influential course. Course and term fixed effects provide a more structured alternative to simple pre/post grade comparisons, but unobserved changes in instructor practice, student composition, institutional policy, or academic support could still contribute to grade differences [21], [23].

For RQ2, the OLSE results show why reliability analysis and CFA are necessary before mechanism modeling. Because OLSE is based on student perceptions of confidence rather than direct behavior, the analysis first evaluated whether the survey responses formed a coherent self-efficacy construct in this dataset [15]–[17]. The similar fit of the five-domain, higher-order, and single-factor CFA models suggests that OLSE operated primarily as a general self-efficacy construct, so domain-specific OLSE interpretations should be treated cautiously. The results support the use of OLSE as a higher-order OLSE Construct in this workflow, but they should not be interpreted as standalone psychometric validation across all settings.

For RQ3, the redesign composite pathways provide a more detailed interpretation than a binary pre/post redesign indicator. The social/vicarious pathway showed the largest indirect estimate, suggesting that interaction, communication, instructor presence, and peer engagement may be especially

relevant for students' confidence in online courses. This interpretation is consistent with online learning research emphasizing interaction, self-efficacy, and teaching/social presence [18]–[20]. Mastery and structural/compliance redesign were also positively associated with OLSE, suggesting that clarity, alignment, accessibility, navigation, and support may reduce uncertainty in online learning. The SEM sensitivity checks using manifest OLSE and overall redesign intensity support the stability of this interpretation under alternative specifications.

C. Practical Use and Limitations

The intended problem owners are instructional design teams, online-learning administrators, and program leaders responsible for evaluating redesign investments. In this setting, LENS is designed to be rerun after each redesign cycle to produce course-level summaries, robustness-checked grade associations, OLSE measurement evidence, and redesign pathway summaries for faculty consultation and program planning. A practical strength of LENS is that it connects three types of evidence that are often evaluated separately: QM-style course review scores, OLSE student-perception survey responses, and final numeric grades. In institutional use, the same pipeline could be updated after each redesign cycle to generate four reusable outputs: linked analytic files, pre/post diagnostic summaries, adjusted grade association estimates, and mechanism-oriented OLSE pathway summaries. These outputs should be used as decision support alongside faculty input, course context, implementation records, and local knowledge [44]–[46].

This study has several limitations. First, the data come from one institution and 19 courses, limiting generalizability. Although the analytic file contains 1,568 student-course observations, redesign variation occurs at the course or course-term level, so the number of courses remains an important inferential constraint. We therefore report CR2/Satterthwaite and leave-one-course-out checks, but the estimates should still be interpreted cautiously. Second, the repeated cross-sectional design does not follow the same students over time, so pre/post differences may reflect variation in student composition as well as redesign implementation. Third, the fixed-effects models do not eliminate possible time-varying confounders, such as changes in instructor practice, grading policies, advising support, student composition, or institutional conditions that may coincide with redesign implementation. Fourth, OLSE is a self-reported confidence measure rather than a direct behavioral trace. Fifth, the high correlations among OLSE domains support a general higher-order OLSE construct but limit domain-specific interpretation. Finally, SEM pathways are interpreted as redesign–OLSE–grade associations rather than causal mediation effects because the observational design does not establish temporal ordering for all constructs at the individual-student level [40].

VI. CONCLUSION

This paper presented LENS, a linked data science workflow for evaluating online course redesign using student-course observations, course-term implementation records, course-level redesign features, OLSE survey responses, and final numeric grades. The workflow addresses a practical institutional analytics problem: how to connect course redesign activities with student outcomes and theoretically meaningful learning mechanisms.

The paper was organized around three research questions. RQ1 examined whether post-redesign offerings were associated with higher grades after accounting for course and term differences. RQ2 evaluated whether OLSE provided a reliable higher-order measurement structure suitable for mechanism analysis. RQ3 tested whether redesign exposure and redesign dimensions were associated with grades indirectly through OLSE. Results showed positive post-redesign grade associations, stability across small-cluster and leave-one-course-out checks, strong OLSE reliability and CFA evidence, successful pre/post measurement comparability checks, single-factor CFA robustness, and positive indirect redesign–OLSE–grade associations across SEM sensitivity specifications.

The main contribution is a transferable applied data-science workflow for linking intervention records, implementation metadata, survey-based latent constructs, and administrative outcomes into a reproducible evaluation pipeline. Rather than treating redesign evaluation as a single pre/post outcome comparison, LENS demonstrates how heterogeneous institutional data layers can be connected to produce interpretable, robustness-checked, decision-support outputs. The workflow is intended to make course redesign evaluation more transparent, reproducible, and interpretable under realistic institutional-data constraints. By linking redesign features, implementation timing, student self-efficacy, and grade outcomes, LENS helps move course redesign evaluation beyond simple pre/post comparisons. When implemented with privacy protections and interpreted as decision support rather than automated decision-making, it offers a practical template for evaluating online course redesign as both an outcome-oriented and mechanism-oriented data science problem [44], [45], [47].

Future implementations of LENS should apply the workflow to larger datasets across additional cohorts, institutions, course formats, and student populations. Larger samples would support stronger modeling of nested course and section structures, including multilevel or Bayesian SEM extensions, while repeated student-level survey and outcome measures would allow future versions of the workflow to examine within-student change over time. Future work should also extend the quantitative findings through qualitative interviews that examine how and why students perceive specific course changes as influencing self-efficacy, supporting framework building for self-efficacy-centered course redesign [48], [49].

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APPENDIX A

SUPPLEMENTARY MATERIAL: OLSE SURVEY DOMAINS

TABLE IX: OLSE survey domains used to measure online learning self-efficacy.

Domain	Survey prompt	Items
Q1	How confident are you that you could do the following tasks in the online course?	8
Q2	How confident are you that you could do the following social interaction tasks with your classmates in the online course?	5
Q3	How confident are you that you could use the following tasks while using online course tools in the course management system?	6
Q4	How confident are you that you could do the following tasks while interacting with your instructor in the online course?	5
Q5	How confident are you that you could do the following tasks while interacting with your classmates in the online course?	6

TABLE X: SEM Sensitivity Checks

Model	Key result
Latent OLSE parcel SEM	Post $\rightarrow$ OLSE $\rightarrow$ grade = 0.157, $p < .001$
Manifest OLSE path model	Post $\rightarrow$ OLSE $\rightarrow$ grade = 0.145, $p < .001$
Redesign-intensity SEM	Intensity $\rightarrow$ OLSE $\rightarrow$ grade = 0.089, $p = .0048$

TABLE XI: OLSE CFA Model Comparison

Model	χ^2	df	p	CFI	TLI	RMSEA	SRMR
Five-domain CFA	442.746	395	.049	0.998	0.998	0.009	0.019
Higher-order CFA	445.796	400	.057	0.998	0.998	0.009	0.019
Single-factor CFA	455.155	405	.043	0.998	0.998	0.009	0.020